

Career Objective

Senior C/C++ Developer with 5+ years of experience in automotive embedded systems development. Strong expertise in modern C++ (C++11/14), STL, data structures, design patterns, and performance optimization on resource-constrained systems. Experienced in unit testing, static analysis (MISRA, Helix QAC), Device Driver and CI/CD practices. Seeking challenging C++ development roles to contribute strong system-level programming and problem-solving skills.

Technical Skills

Programming Languages : Embedded C, C++11/14/17, Python

C++ Concepts : Data Structures, Templates, Design Patterns, STL, Basic Multithreading

Operating Systems : RTOS, Linux, FreeRTOS

Testing & Quality Tools : Google Test, Coverity Static Analysis, Helix QAC, MISRA

Microcontroller : 16/32bit Microcontrollers from Renesas RL78, STM32, Aurix

Configuration Tools : IAR Embedded Workbench, IBM DOORS, Git, IBM DNG, GitLab, CANoe, WinIdea, CubeSuite, Visual Studio, Cmake, Bazel, Coverity

Bug Tracking Tools : JIRA, PTC, ALM

Communication Protocol : CAN, UDS, SPI, I2C

AI Tools : Microsoft Copilot, GitHub Copilot, GitLab AI Code Review

Professional Experience

Forvia Hella, Pune

(Duration: Feb 2023 to Present)

1. RLS Automotive Lighting Control System

Responsibilities:

- Developed and optimized RTOS based application-level logic using C++11/14, interfacing with hardware drivers.
- Implemented diagnostic services using C++11/14 to support vehicle health monitoring, fault detection, and ECU diagnostics.
- Refactored existing Embedded C Application layer, MCAL modules, into C++ architecture using OOP concepts, enhancing scalability and readability.
- Applied data structures and architectural design patterns to improve system performance.
- Supported board bring-up activities including hardware initialization, peripheral validation, and startup debugging.
- Performed unit testing using Google Test (GTest) and ensured code quality through static analysis (Helix QAC) with MISRA compliance.
- Developed a C++11-based SWC generator tool to automatically generate SWC code.
- Utilized AI-assisted tools such as Microsoft Copilot and GitHub Copilot to improve coding efficiency, perform code reviews, and reduce development time.
- Experienced in Waterfall and Agile Development methodologies, including Scrum, Sprint Planning, Reviews, and Retrospectives.
- Extensive use of version control tools (Git) and ALM, PTC for efficient development tracking.
- Applied Standard Template Library (STL containers/algorithms). Basic understanding of multithreading concepts and thread-safety fundamentals.
- Hands on experience of C++ memory management, including pointers, and smart pointer.

Hardware Platform: 16-bit RL78/F12

Tools Used: C++11/14, IAR Workbench, CubeSuite CA/CX, Applilet3, UDS, Git, ALM, IBM DNG, Design Pattern, Google Test

2. Zonal Controller(BCM)**Responsibilities:**

- Configured and integrated AUTOSAR MCAL modules including ADC, DIO, SPI, PWM, and Port, enabling communication between application software and microcontroller hardware.
- Performed module-level testing of MCAL components based on system requirements.
- Configured OS tasks and integrated multicore and Kernel support for MCAL modules.
- Worked in ASPICE-compliant development environment with exposure to Functional Safety (ISO 26262) practices.
- Worked on the SPI as slave mode at register level also worked on circular queue to handle the data.
- Worked on the RTE layer to configure communication between BSW and ASW.
- Performed runtime analysis to calculate runnable, core, and task utilization.
- Created design documentation for software modules using IBM Rhapsody.
- Created and maintained software requirements and traceability using IBM DNG.

Hardware Platform: RH850-U2B**Tools Used:** EC++11/14, DaVinci Configurator, DaVinci Developer, CANoe, WinIdea, Git, IBM DOORS, JIRA, Bazel, Coverity, Astree, UDS, CAN, GTest**Previous Experience: Amdocs, Pune****(Duration: March 2022 – Feb 2023)****1. ADAS Analysis Test Framework****Client:** JLR**Responsibilities:**

- Designed and implemented C/C++ modules within the ADAS Analysis Framework (RESIM-SIL) to model Radar SIL for SRR5 and Radar ECU, ensuring seamless integration with both SIL and HiL test environments.
- Developed a framework for vehicle log analysis, running Feature Functions (FF) like Blind Spot Detection and Lane Departure Warning.
- Collaborated with cross-functional teams, Devops and Testing, to enhance system performance.
- Ensured code quality and optimized performance by following C++ best practices.
- Ensured the accuracy and integrity of radar data and logs before hardware/software delivery.

Tools Used: C++, SQL, VS Code, Python, Valgrind, GitLab, JIRA, SonarQube, Linux, vi.**Previous Experience: Capgemini Engineering, Bengaluru (Duration: Jan 2021 – March 2022)****1. Front Radar****Client:** Volvo**Responsibilities:**

- Written CAN/CAN-FD stack for Communication on top of the MCAL Layer
- Configured DCM, DEM and MCAL modules in AUTOSAR as per UDS protocol and customer specifications.
- Implemented DIDs, RIDs and DTCs (Diagnostic Trouble Codes) in software components using Embedded C and validated them.
- Developed a Python script to automate the deletion of ports and runnables from SWC.
- Experience in configuring and integrating AUTOSAR MCAL modules including ADC, DIO, SPI, PWM, and Port, enabling communication between application software and microcontroller hardware.
- Developed unit test cases in C++ using Google Test (GTest) and Google Mock to validate Diagnostic Stack (UDS services) functionality

Tools Used: C++11/14, DaVinci Configurator, DaVinci Developer, CANoe, WinIdea, Git, IBM DOORS, JIRA, Helix QAC, Code Collaborator, MISRA, UDS, CAN, GTest.

Akash PatilEMAIL-ID: akashpatil199801@gmail.com

Academic and Professional Qualifications

Qualification	Institution	Year	CGPA/Percentage
B.E	BMS Institute of Technology (BMSIT&M)	2020	8.06
Class 12	KLES PU College	2016	89.83%
High School	Sri Mahakali Comp College	2014	92%

Personal Information

Name : Akash Patil
Gender : Male
Languages Known : English, Hindi, Kannada and Marathi.
Date of Birth : 21st Feb 1998
Mobile No. : +91 9901068018

Declaration

I hereby declare that the above information is true to the best of my knowledge and belief.

Place:

Date:

(Akash Patil)

Akash Patil

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